MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP BCA (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER **MG2SECBCA100** WEB TECHNOLOGIES

SCHEME FOR EVALUATION

QP Code :**25900044**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skills(S), Interest(I), and Appreciation(Ap)

Duration : 1 Hour Max Marks: 35

Part A

Short Answer questions. Answer any 4 questions.

Each question carries 5 marks

1. Why is DNS important for internet browsing? [K] / [CO1]

Internet understands IP address only. The domain name to ip address conversion is done by DNS server. ---**explanation 5 marks**

2. Describe how HTTP and FTP work and their roles in web communication. [U] / [CO1]

HTTP is used for browsing websites, while FTP is used for transferring files to/from a server.
http – 2.5 marks
ftp – 2.5 marks

3. How would you set up a basic web server on Linux? List the steps. [A] / [CO1]

Explanation – 3 marks ..steps 2 marks

Steps to set up a basic web server on Linux:

i. Install a web server software like Apache.
ii. Start the web server so it begins running.
iii. Set it to start automatically when the computer turns on.
iv. Place your website files in the default web directory: /var/www/html.
v. Finally, open a browser and check if the server is working by typing the server's IP address or http://localhost.

4. Describe the use of input box in JavaScript. [U] / [CO2]

Explanation– 3 mark
Example code – 2 marks

5. Implement a function that removes duplicate values from an array. [A] / [CO2]

Proper function definition - 3 marks
Logic - 2 marks

```
function removeDuplicates(arr)
{  var unique = [];
        var isDuplicate;
        for (var i = 0; i < arr.length; i++)
      {    isDuplicate = false;
              // Check if arr[i] already exists in unique[]
              for (var j = 0; j < unique.length; j++)
            {       if (arr[i] === unique[j]) {    isDuplicate = true;  break;  }     }
```

```
                    // If not duplicate, add to unique[]
                    if (!isDuplicate) {     unique[unique.length] = arr[i];    }
                }
                return unique;
            }
```

6. Discuss the disadvantages of AJAX and how they can impact web development. [U] / [CO2]

AJAX allows web pages to update dynamically without reloading. This enhances user experience by enabling faster interactions. The main concern while using AJAX in web development are compactability issues and security compromises. Security concerns and the potential for increased server load are other drawbacks that may affect performance and require additional development efforts.

Discussion with disadvantages- 4 marks .
Impact on web development – 1 mark

[5 X 4 = 20]

Part B

Essay questions. Answer any 1 question.
Each question carries 15 marks

7. Develop a well-structured HTML page that includes multiple header levels, paragraphs, and line breaks. Explain the role of each tag used. [A] / [CO1]

10 marks for the html code and 5 mark for explanations - header tags, paragraph and break.

i. **`h1> to <h6>`**: These are heading tags used to define headings of different importance. `<h1>` is the main heading, and `<h2>` to `<h6>` are progressively smaller headings, used for organizing content into sections and subsections.
ii. **`<p>`**: Defines a paragraph. Text inside this tag is grouped together as a block of text.
iii. **`<br>`**: Inserts a line break, causing the content that follows to start on a new line.

8. Define JavaScript functions, events, and DOM Traversing. Provide examples for each concept. [U] / [CO2]

5 marks each for each concept.
2 mark for explanation and 3 mark for code.

A **function** in JavaScript is a block of reusable code that performs a specific task when called.

```
<html> <head>   <title>Alert Function Example</title> </head>
<body>
  <button onclick="hello()">Click Me</button>
  <script>
    function hello() {       alert("Hello World ! Welcome to JavaScript");     }
  </script> </body> </html>
```

An **event** is an action that occurs in the browser, such as a click, mouseover, or key press. JavaScript can respond to these events.

```
<button onclick="showMessage()">Click Me</button>
```

DOM traversing means navigating through the structure of an HTML document (called the Document Object Model or DOM) to access, modify, or interact with specific elements using JavaScript. It allows developers to move between parent, child, and sibling nodes in the HTML

tree. This is done using properties like parentNode, childNodes, firstElementChild, nextElementSibling, etc.

```
<html> <body> <div id="Parentdiv">   <p id="ChildPara1">This is inside a div.</p>
</div> <script>
  var innerPara = document.getElementById("ChildPara1");    // Get the <p> element
  var parentDiv = innerPara.parentNode;                 // Traverse to parent <div>
  alert("Parent ID is: " + parentDiv.id);              // Show the parent ID
</script> </body></html>
```

[15 X 1= 15]